circuit. Install PCScope.exe program (developed by author) in your Windows PC and open the application. Next, open the Arduino sketch from Arduino IDE and compile the sketch. Connect the Arduino board to the PC and flash the sketch into the microcontroller on the Arduino board.

The ADC of Arduino can measure voltages up to 5V. So it is advisable to add a small protection circuit to limit the input voltage to 5V and clamp the negative voltage. A low-power, fast-switching diode like 1N4148 can be used to protect the input pin. Connect a 10-kilo-ohm resistor in series with the input. It will work as a current limiter in case the input goes beyond 5V. Additional voltage dividers can be used in case you need to measure voltages higher than 5V.

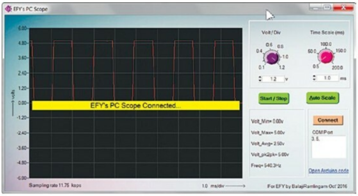

Fig. 3: Message after the hardware successfully connects to the PC

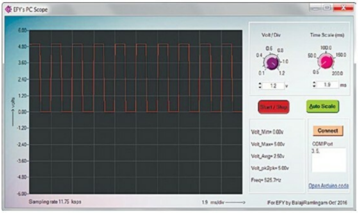

Fig. 4: Test signal of 525Hz square waveform captured on the screen

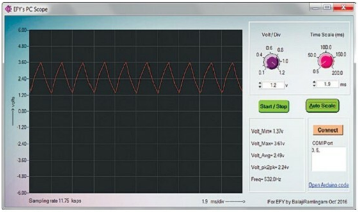

Fig. 5: Test signal of 530Hz triangular waveform captured on the screen

Software

Arduino sketch. The sampling rate of this PC scope application is limited by the rate at which the data is sent to the PC. Baud rate of 115000 gives time interval of around 85 µs. It is important to get the ADC signals much before this time to get reliable data plotting. The sketch reads pin A0 of Board1 and sends to UART at 115200 baud rate. At this speed, bytes of the input are pushed at time intervals of around 85µs.

By default, the ADC configuration of the Arduino gives samples every 116µs. So here the ADC is configured with additional lines of code to get samples faster than 85µs by setting the prescaler to 16. With this, you get ADC conversion every 20µs, which is much faster than the UART data transfer rate.

PC software. As stated earlier, the front-end PC software for signal acquisition and processing is developed using NI LabWindows. The serial port data is captured through Arduino at regular time intervals and plotted as a graph on the screen using the Plot function library. The display points along X-axis are calculated based on the user-defined time scale. The Y-axis range is set using the voltage selection control.

Testing

After installing the PC scope application, click 'Connect' button on your PC screen to connect to the Arduino board (Fig. 2). When the board gets connected to your PC, you will get a confirmation message for three seconds as shown in Fig. 3.

Feed any squarewave input of up to 5kHz at CON1. The software must plot its output waveform on your PC. Square and triangular output waveforms of 525Hz and 530Hz captured on the screen during testing are shown in Figs 4 and 5, respectively. Similarly, you can feed rectangular or pulse inputs (but not sine waves) to get output waveforms. EFY

EFY Note
The source code of this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*

The author is a program manager at Robert Bosch, Bengaluru. He has filed several patents in automotive electronics and published papers in SAE conferences and several international magazines